

AGRILINK: DIRECT FARM-TO-COMMUNITY SUPPLY PLATFORM

Makerspace Project Handover & Interdisciplinary Student Invitation Guide

Open for Students from All Disciplines | Academic Year 2026

PROJECT OPPORTUNITY & INVITATION SUMMARY

Welcome to the AgriLink Project: AgriLink connects local farmers directly to retail customers, university hostels, hospitals, and local restaurants. Whether you are interested in software, electronics, business strategy, user design, photography, logistics, or biology—you do not need to be a programmer or engineer to contribute. This document explains what we are doing, what we have built so far, and how you can get involved.

1 Project Title

Project Name: AgriLink — Smart Farm-to-Customer Supply & Recommendation Platform

Core Focus: Connecting rural farmers and agricultural aggregator hubs directly to consumers, cutting out unfair middlemen, reducing food waste, and making fresh farm produce affordable and transparent.

2 The Problem

Our current agricultural supply chain has several major real-world challenges:

- **Too Many Middlemen:** Produce passes through 4 to 6 intermediaries before reaching consumers. As a result, farmers receive only 20–30% of the final price, while consumers pay high prices.
- **High Food Wastage:** Because of delays, lack of cold-chain monitoring, and poor demand planning, 25% to 35% of harvested fruits and vegetables spoil before they reach buyers.
- **Lack of Information:** Farmers rarely know what local institutions (hostels, hospitals, restaurants) need on a daily basis, and buyers have no way of knowing how fresh their food actually is or where it came from.

3 Why Does This Matter?

Agriculture supports more than half of the country's population, yet farming families face high economic uncertainty. Working on this project allows students to:

- **Create Real-World Social Impact:** Fair pricing directly improves farmer livelihoods and lowers food costs for student messes and local families.
- **Reduce Environmental Waste:** Better matching of supply and demand prevents nutritious food from ending up in landfills.
- **Gain Multi-Disciplinary Experience:** Work with peers across departments on practical challenges that blend technology, market economics, design, and rural operations.

4 Our Idea / Proposed Solution

AgriLink is an open, easy-to-use digital ecosystem that connects all stakeholders in one transparent loop:

1. **For Farmers:** A simple portal to list their harvested crops, view fair market prices, and receive direct bulk orders from nearby buyers.
2. **For Rural Aggregator Hubs:** Facilities where smallholder crops are gathered, checked for quality, safely packed, and prepared for dispatch.
3. **For Customers & Institutions:** An intuitive app/website to browse fresh produce, view harvest freshness, join group orders (bulk buying for hostels/apartments), and schedule recurring meal deliveries.
4. **Smart Matching:** Recommending specific nearby harvest batches based on harvest date, distance, quality grade, and buyer preferences.

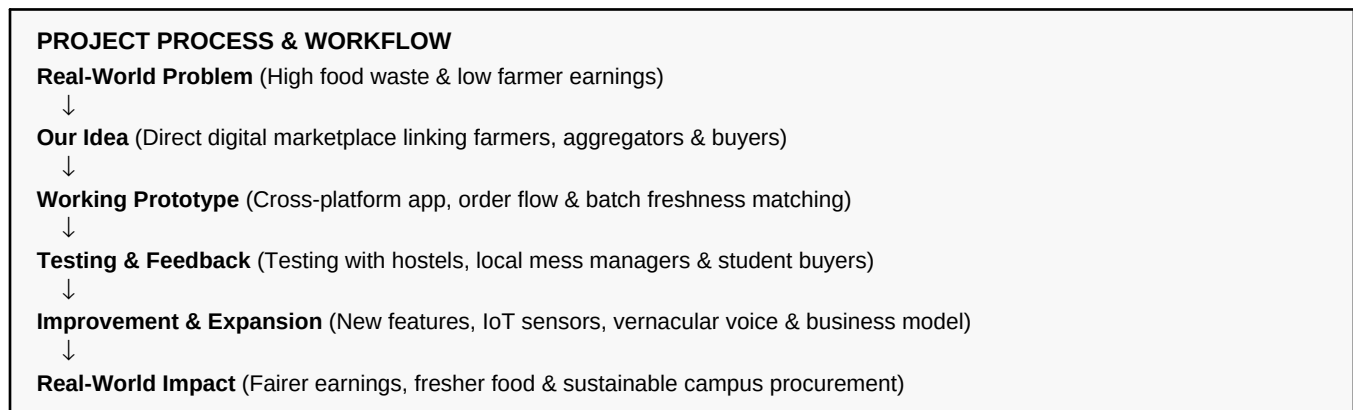
5 What Do We Need?

The project brings together different kinds of resources and expertise across five areas:

Area	What We Need	Examples in the Project
Software & Digital	Applications, databases, and analysis tools	User apps, order management, smart search, and recommendation feeds.
Hardware & Devices	Sensors, development boards, and components	Temperature/freshness crate trackers, digital weighing scales, and inspection cameras.
Data & Information	Datasets, surveys, and real-world numbers	Mandi crop prices, hostel food consumption logs, buyer preferences, and harvest cycles.
Design & Creative	Visuals, interfaces, videos, and branding	Clean mobile screens, user journey maps, instructional posters, and explainer videos.
Research & Business	Reports, business plans, and user studies	Cost-benefit analysis, farmer interviews, logistics planning, and policy reviews.

6 How Does It Work?

At a high level, the project follows a clear, continuous development lifecycle:



7 Where Are We Now?

Current Stage Overview: Idea → [**Working Prototype**] → Testing → Campus Pilot → Deployment

Stage	Current Status	What This Means
Core Prototype	[COMPLETED]	The foundational system is fully built and working on test data.
User Testing	[IN PROGRESS]	Gathering user experience feedback and verifying ordering workflows.
New Features	[OPEN FOR STUDENTS]	Many new extensions in software, design, hardware, and business are ready to be built.

8 What Have We Built?

The team has already constructed a functioning system prototype containing:

- **User-Friendly Marketplace App:** A clean interface where buyers can sign in, search for vegetables, fruits, and grains, filter by freshness and location, and place orders.
- **Farmer & Aggregator Portals:** Screens for farmers to upload available harvest quantities, set prices, and see incoming requests.
- **Smart Recommendation Engine:** Automatically suggests best-fit crop listings based on proximity, harvest freshness percentage, and buyer preferences.
- **Live Verification Test:** An automated test suite demonstrating sub-second response times across all services.

9 What Has Been Completed?

Completed Work Items	Currently Active Work Items
[x] Initial problem definition & user research [x] Database structure with 100+ sample crop records [x] Core marketplace application & role logins [x] Recommendation model comparison & evaluation [x] Comprehensive technical report & architecture documentation	[] Designing new user interfaces & onboarding graphics [] Collecting campus hostel & canteen buying requirements [] Testing physical crate sensors in the makerspace [] Creating pitch decks and competition demonstration videos [] Expanding regional language voice support for farmers

10 What Can Students Work On Next?

We have structured upcoming tasks by experience level so everyone can start comfortably:

Skill Level	Exciting Tasks & Projects You Can Take Up
Beginner-Friendly (No coding required)	• Interview campus hostel/mess managers about their vegetable ordering needs. • Conduct user surveys on mobile app usability and ease of navigation. • Design posters, branding assets, and project presentation slide decks. • Test the current mobile app and record usability feedback or bug reports. • Create short demonstration videos and visual guides for new users.
Intermediate (Basic tools / design / coding)	• Redesign mobile app screens in Figma / Flutter for a smoother visual experience. • Analyze historical mandi market price trends and create visual dashboards. • Prototype a simple temperature & humidity sensor node in the makerspace. • Build a group-buying / split-bill feature for student hostel roommates. • Write user stories and documentation for campus competition submissions.
Advanced (Specialized / technical / research)	• Build a smartphone camera scanner to automatically detect fruit ripeness / defects. • Develop an intelligent route optimizer for multi-stop delivery vehicles. • Integrate local voice-to-text (vernacular speech) so farmers can speak to list crops. • Implement blockchain / QR batch tracking for pesticide audit trails. • Author research papers for student symposiums and international conferences.

11 Who Can Contribute?

A MULTI-DISCIPLINARY INVITATION

Important Principle: You do not need to be a computer science student or have coding experience to make a valuable contribution. Real projects need diverse minds: researchers, designers, business analysts, domain enthusiasts, communicators, and builders alike.

Three Broad Ways to Contribute:

Track	Interest Areas	Example Contributions
1. Technology & Engineering	Software, Data, Electronics, Testing, Hardware	App features, database queries, sensors, automation scripts, and test cases.
2. Design & Communication	UI/UX, Visual Design, Video, Writing, Social Media	Figma prototypes, user journey diagrams, video demos, and presentations.
3. Business, Research & Society	Economics, Finance, Surveys, Sustainability, Policy	Cost-benefit models, user research surveys, market analysis, and case studies.

Contribution Guide by Academic Background:

Student Background / Major	How You Can Contribute to AgriLink
Computer Science / IT	Build frontend features, backend APIs, mobile screens, and database optimizations.
Data Science / AI	Analyze crop datasets, improve recommendation algorithms, and train price predictors.
Electronics / ECE / EEE	Work with ESP32 sensor boards, temperature/humidity monitors, and crate GPS units.
Mechanical / Mechatronics	Design lightweight, shockproof, and ventilated produce crates and mounting mechanisms.
Design / Media / Animation	Create intuitive mobile interfaces, graphic posters, explainer videos, and branding.
Business / Management	Develop business models, study customer adoption, and create commercialization plans.

Finance / Economics	Conduct cost-benefit analysis, supply-demand pricing studies, and margin comparisons.
Biotechnology / Agriculture	Provide domain guidance on crop shelf life, organic certifications, and storage guidelines.
Humanities / Psychology	Conduct farmer usability studies, buyer behavior surveys, and social impact evaluations.
Any Discipline / First Years	Project management, documentation, testing, user feedback collection, and creative brainstorming.

12 How Can This Connect to Courses?

Students can use AgriLink work to earn course credits, mini-projects, assignments, and capstone milestones:

Curriculum Area	Example Project / Assignment Deliverable
Programming & Web / Mobile	Implement a new user screen, cart checkout, or notifications system.
Data Analysis & Statistics	Perform statistical analysis on buyer trends and seasonal crop demand.
Hardware & IoT Labs	Assemble and calibrate an environmental monitoring crate module.
Human-Computer Interaction (HCI)	Conduct heuristic evaluation and usability testing on farmer interfaces.
Supply Chain & Logistics	Model aggregator collection routes and compute transportation cost savings.
Marketing & Entrepreneurship	Develop a go-to-market plan for campus hostels and local neighborhood clusters.
Technical Writing & Research	Prepare comprehensive project dissertations, research papers, or user manuals.

13 Competitions, Research & Other Opportunities

Being part of AgriLink gives you a strong foundation to compete and showcase your achievements:

- **Smart India Hackathon (SIH):** Fits national problem statements on smart agriculture, rural supply chains, and farmer welfare.
- **Innovation & Sustainability Contests:** Ready for AgriThon, UN SDG Youth Challenges, and campus entrepreneurship competitions.
- **Research Publications:** Empirical benchmark results and user studies can be developed into conference papers (IEEE/ACM/Springer).
- **Campus Startup Opportunities:** Potential to incubate the platform as a real campus-to-farm procurement service.

14 Current Team & Mentors

Role	Name / Department	Area of Guidance / Collaboration
Faculty Guide	Project Coordinator / Faculty Mentor	Overall academic oversight and institutional coordination
Makerspace Mentor	Innovation Lab Coordinator	Hardware tools, workspace access, and rapid prototyping
Senior Student Leads	AgriLink Founding Student Developers	Architecture onboarding, peer mentorship, and codebase support
Open Student Roles	Open for 1st, 2nd, 3rd & 4th year students	Development, research, design, business, and testing tracks

15 What Is Already Available?

You don't need to start from scratch. The project already provides comprehensive resources:

- **Equipment & Tools:** ESP32 boards, sensors (temperature, humidity, soil), load cells, and 3D printing equipment in the makerspace.
- **Software & Platforms:** Fully configured development repository, working mobile app, API backend, and cloud database.
- **Knowledge & Data:** Clean datasets of crop varieties, price histories, synthetic customer interactions, and full research reports.
- **Workspace & Community:** Dedicated makerspace lab tables, collaborative discussion channels, and regular peer check-ins.

16 Project Files & Resources

All project assets are neatly organized and easy to navigate:

Resource	Location / Folder	Contents & Purpose
Application Frontend	/lib/	All Flutter mobile and web screens, UI themes, and models.
API Backend	/fastapi_backend/	FastAPI server handling listings, orders, and recommendations.
Data & Models	/ML_Benchmark/	Datasets, model benchmarking code, and performance charts.
Database Schemas	/Backend/database/	SQL migration files and 100+ sample seed data entries.
Academic Report	/Academic_Project_Report/	Detailed project report docx and reference documentation.

17 How Can You Get Started?

4-STEP EASY ONBOARDING PROCESS

Step 1: Understand the Project: Read through this guide to see the big picture and identify what interests you.



Step 2: Choose Your Area of Interest: Pick an area you'd like to explore (e.g., Design, Surveys, App Features, Sensors, or Business).



Step 3: Meet the Team & Mentors: Visit the makerspace or reach out to the project leads to discuss your chosen track.



Step 4: Start with a Small, Guided Task: Begin with an easy starter activity to gain confidence and gradually take on bigger challenges.

Note: No prior experience is required for many project activities. We will guide you step by step!

18 Why Join This Project?

Joining AgriLink offers significant personal and professional value for your college journey:

- **Practical Portfolio Experience:** Build tangible artifacts (designs, hardware, software, or market studies) to showcase on your CV and LinkedIn.
- **Cross-Disciplinary Teamwork:** Learn how to collaborate effectively with students outside your immediate branch or year.
- **Hackathon & Competition Mentorship:** Receive active mentoring from seniors and faculty for prestigious national competitions.
- **Leadership & Initiative:** Take ownership of independent sub-tracks and lead project milestones.

19 Photos & Demonstration Visuals

Reserved blank visual layout sections for team photos, application screenshots, hardware prototypes, and demo links:

[VISUAL 1: MOBILE & WEB APPLICATION INTERFACE]

(Blank area reserved for visuals: Recommended size: 500 x 180 px)

Paste prototype screenshots, hardware photos, fieldwork pictures, or demo QR codes here.

[VISUAL 2: MAKERSPACE HARDWARE PROTOTYPE / FIELDWORK PHOTOS]

(Blank area reserved for visuals: Recommended size: 500 x 180 px)

Paste prototype screenshots, hardware photos, fieldwork pictures, or demo QR codes here.

[VISUAL 3: PROJECT DEMONSTRATION VIDEO QR CODE / POSTER]

(Blank area reserved for visuals: Recommended size: 500 x 150 px)

Paste prototype screenshots, hardware photos, fieldwork pictures, or demo QR codes here.

JOIN THE AGRILINK TEAM TODAY

Ready to Join? Reach out to the Faculty Project Guide or drop by the Makerspace Innovation Lab. Every contribution—big or small, technical or creative—makes a real difference in empowering local farming communities!